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Practical 10

Aim: Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. Write a Program to calculate the total head movement (in number of cylinders) incurred while servicing these requests.

Name:

USN:

Semester / Year:

Academic Session:

4th SEM / 2nd YEAR
2025-26

Date of Performance:

Date of Submission:

❖ CODE :

```
GNU nano 8.7
#include <stdio.h>
#include <stdlib.h>
#include <limits.h>

int main() {
    int request[8] = {98, 183, 41, 122, 14, 124, 65, 67};
    int visited[8] = {0};
    int head = 53;
    int total = 0;
    int n = 8;

    for (int i = 0; i < n; i++) {
        int min = INT_MAX;
        int index = -1;

        for (int j = 0; j < n; j++) {
            if (visited[j] == 0) {
                int distance = abs(head - request[j]);

                if (distance < min) {
                    min = distance;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        printf("Head moves from %d to %d with seek %d\n", head, request[index], min);

        total += min;
        head = request[index];
    }

    printf("\nTotal Head Movement = %d\n", total);

    return 0;
}
```

❖ OUTPUT :

```
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ gcc SSTF.c -o SSTF  
  
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ ./SSTF  
Head moves from 53 to 41 with seek 12  
Head moves from 41 to 65 with seek 24  
Head moves from 65 to 67 with seek 2  
Head moves from 67 to 98 with seek 31  
Head moves from 98 to 122 with seek 24  
Head moves from 122 to 124 with seek 2  
Head moves from 124 to 183 with seek 59  
Head moves from 183 to 14 with seek 169  
  
Total Head Movement = 323
```